MELBA Journal Machine Learning for Biomedical Imaging

MICCAI Open Data 2026 x MELBA sample article

Firstname1 **Name1** [1,3], Firstname2 Name2 [2,3]

1 Melba journal, The Internet, CERN, CH
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Abstract

This MELBA Resource track is dedicated to resources that can facilitate i) reproducible research within the field of biomedical imaging and ii) fair benchmarking of machine learning algorithms. MELBA Resource publications are articles systematically describing publicly available resources, such as i) research datasets and ii) open-source software tools, that reside at the intersection of biomedical imaging and machine learning. Our aim is to enhance the sharing and reuse of resources, encourage their broader dissemination and repurposing, as well as acknowledge those who contribute by sharing.

Keywords

Melba journal, resource track, open data

Article informations

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1. Background

This section should provide the background and motivation for the specific "resource".

The gap/need in the literature that makes the presented "resource" valuable for ML/AI projects should be indicated.

Reference to similar "resources", further highlighting the unaddressed gap or complementing the current "resource", should also be reported here.

2. Summary

This section should summarise the vision, mission, scope, and objectives of the "resource".

Who is the target audience? Who does it serve?

The readiness of the "resource" for use in a medical imaging ML/AI project is expected to be highlighted here. For example, is the presented dataset AI-ready? Does use of the presented tool require programming knowledge? Is the presented tool executable in a low-/zero-code principle?

3. Discussion

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Table 1: By convention, Table caption goes on top.

Left	center	right
111	222	333
444	555	666

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Use \cite{} for reference that is part of the sentence, and \citep{} for references in parenthesis. For example, Viola and Wells III (1997) is awesome. Also, this is a citation (Viola and Wells III, 1997).

4. Resource Availability

4.1 Summary statement

This section should summarise the importance of the resource (e.g., new datasets for underserved populations), in not more than 4 sentences.

4.2 Data/Code Location

Specific link should be added here.

Examples of repositories are given at the Public Repositories section below.

4.3 Potential Use Cases

This section should indicate the potential ML/imaging use cases that the authors envision their resource facilitating and the specific potential clinical domain(s).

4.4 Licensing

Clearly describe the licensing terms.

If there is an existing licence that is followed this could just be a single sentence.

4.5 Ethical Considerations

Language referring to associated documentation, e.g., IRB, Informed Consent.

For human data collection this section must describe the consent or opt-in/out process.

5. Methods

This section should highlight either (for 'Data Resource' papers) all methods/techniques/pipelines that the authors used to prepare the presented data, or (for 'Tools Resource' papers) that were included in the presented tool.

5.1 Data Details

This section should indicate inclusion criteria, as well as describe any associated demographic, clinical, molecular, and other related details.

5.2 Methods Used for the Data Creation

This section is only relevant for 'Data Resource' papers.

All methods and equipment that were used from data acquisition to final processing should be described in detail here.

Describe the AI-ready state of the dataset. Is it ready to be ingested by a DL algorithm? If not, what are the potential steps that one needs to take for this?

5.3 Software stack

This section is only relevant for 'Tools Resource' papers.

A textual description and a graphical illustration must be included here.

5.4 Flowchart(s) illustrating the tool's process flow

This section is only relevant for 'Tools Resource' papers.

For example, this flowchart could be indicative of the ML training procedure in this tool.

This flowchart does not necessarily need to be a formal Data Flow Diagram

5.5 Functionalities (For Tools Resource papers only)

This section is only relevant for 'Tools Resource' papers.

6. Validation

6.1 Description of approaches ensuring quality of data

This section is only relevant for 'Data Resource' papers.

Subsections included here should be corresponding to the "Methods Used for the Data Creation" subsection above.

6.2 Experimental Design & Results

This section is only relevant for 'Tools Resource' papers.

Indications of the reproducibility of the code.

CI/CD - Unit Test code coverage

Example data from published or unpublished studies

7. Acknowledgements

Funding and conflicts of interest.

References

Paul Viola and William M Wells III. Alignment by maximization of mutual information. *International journal of computer vision*, 24(2):137–154, 1997.

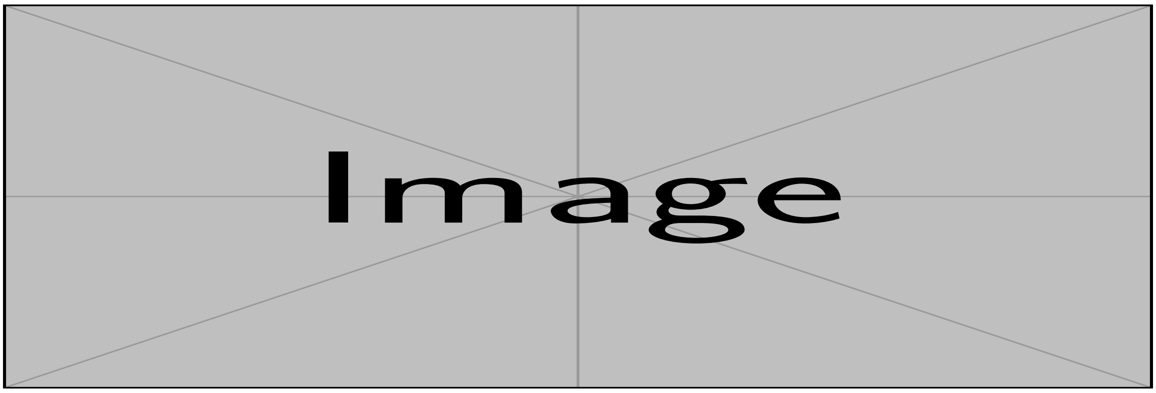

Figure 1: Example figure spanning the two columns.

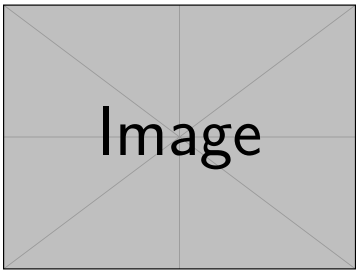

Figure 2: Example figure. Notice that the caption goes below.

Appendix A. Methods

A.1 Equations

We estimate the deformable template parameters θ_t and the deformation fields for every data point using maximum likelihood. Letting $\mathcal{V} = \{\boldsymbol{v}_i\}$ and $\mathcal{A} = \{a_i\}$,

$$\begin{aligned}\hat{\theta}_t, \hat{\mathcal{V}} &= \arg\max_{\theta_t, \mathcal{V}} \log p_{\theta_t}(\mathcal{V}|\mathcal{X}, \mathcal{A}) \\ &= \arg\max_{\theta_t, \mathcal{V}} \log p_{\theta_t}(\mathcal{X}|\mathcal{V}; \mathcal{A}) + \log p(\mathcal{V}), \qquad (1)\end{aligned}$$

where the first term captures the likelihood of the data and deformations, and the second term controls a prior over the deformation fields.

Proof Awesome proof. ■

A.2 Long equations in two columns

Note that equations can fill the width pretty quickly when there are two columns. Different mathemtical environments, beyond the basic `\begin{equation}` may be of use. Taking as an example the binomial formula which overflows in Eq. (2):

$$(1+x)^n = \underbrace{(1+x) \times ... \times (1+x)}_{n \text{ times}} = \sum_{k=0}^{n} \binom{n}{k} x^k = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} x^k. \qquad (2)$$

The `\begin{multline}` environment is an easy although crude fix:

$$\begin{multline}(1+x)^n = \underbrace{(1+x) \times ... \times (1+x)}_{n \text{ times}} \\ = \sum_{k=0}^{n} \binom{n}{k} x^k \\ = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} x^k. \qquad (3)\end{multline}$$

`\begin{align}` tends to give the most control over the final result:

$$\begin{aligned}(1+x)^n &= \underbrace{(1+x) \times ... \times (1+x)}_{n \text{ times}} \\ &= \sum_{k=0}^{n} \binom{n}{k} x^k \\ &= \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} x^k. \qquad (4)\end{aligned}$$

A.3 Math styles

Different font styles can be used for equations:

- `$a b c A B C 1 2 3$`: $abc ABC 123$
- `$\mathbf{a b c A B C 1 2 3}$`: $\mathbf{abcABC123}$
- `$\mathfrak{a b c A B C 1 2 3}$`: $\mathfrak{abcABC123}$
- `$\mathcal{ABC}$`: $\mathcal{ABC}$
- `$\mathbb{ABC}$`: $\mathbb{ABC}$
- `$\mathsf{ABC}$`: ABC
- `$\mathtt{ABC}$`: $\mathtt{ABC}$

Text and names in equations should be dealt with the `\text` command, for instance: `$\mathcal L_{\text{SuperLoss}}$`: $\mathcal{L}_{\text{SuperLoss}}$ and not $\mathcal{L}_{SuperLoss}$.

Appendix B. Appendix section

B.1 Appendix subsection

B.1.1 Appendix subsubsection

Appendix paragraph Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

In `itemize` and `enumeration`, longer sentences and text in paragraphs could require manual hyphenation with `\-` in case of boxes under- or over-flows:

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B.2 Revision

We provide, in `melba.sty` a helpful command to color modifications after a revision: `\revision{}`. It is automatically de-activated for papers compiled with the `accepted`, `arxiv` or `specialissue` options.

B.2.1 It can also color whole sections and paragraphs

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